



4.7 The chi-squared test

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The chi-squared test of independence assesses whether two categorical variables are associated, using a contingency table of observed frequencies O . Expected frequencies are $E = (\text{row total} \times \text{column total}) / \text{grand total}$, and the test statistic is $\chi^2 = \sum (O - E)^2 / E$, compared against a critical value from the χ^2 distribution with degrees of freedom $(\text{rows} - 1)(\text{columns} - 1)$ at a chosen significance level (or its p -value from technology). If $\chi^2(\text{calculated}) > \text{critical value}$ (equivalently, $p < \text{significance level}$), H_0 (independence) is rejected.

Worked example: follow the reasoning

A survey of 200 people classifies them by gender (M/F) and preferred drink (Tea/Coffee): M-Tea 40, M-Coffee 60, F-Tea 50, F-Coffee 50. Test at the 5% level whether drink preference is independent of gender ($df=1$, critical value=3.841).

Solution: Row totals $M=100$, $F=100$; column totals $\text{Tea}=90$, $\text{Coffee}=110$. Expected frequencies: $E(M, \text{Tea})=45$, $E(M, \text{Coffee})=55$, $E(F, \text{Tea})=45$, $E(F, \text{Coffee})=55$.

$\chi^2 = (40 - 45)^2 / 45 + (60 - 55)^2 / 55 + (50 - 45)^2 / 45 + (50 - 55)^2 / 55 \approx 2.02$. Since $2.02 < 3.841$, do not reject H_0 : no significant evidence that drink preference depends on gender.

Quick check · multiple choice

For a 3×4 contingency table, what are the degrees of freedom for a chi-squared test of independence?

- A 12
- B 7
- C 6
- D 3

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A researcher tests for an association between exercise frequency (Low/High) and diet type (Vegetarian/Non-vegetarian) among 180 people: Low-Veg 25, Low-NonVeg 65, High-Veg 35, High-NonVeg 55. (a) State the null and alternative hypotheses. (b) Calculate the expected frequency for the High-Veg cell. (c) Calculate the chi-squared test statistic. (d) Given the critical value at 5% significance with 1 degree of freedom is 3.841, state a conclusion. [8]

Final answer: _____



Answers and explanation

Question 1 [8]

Row totals: Low=90, High=90; column totals: Veg=60, NonVeg=120. (a) H_0 : exercise frequency and diet type are independent. H_1 : they are not independent (associated). (b) $E(\text{High}, \text{Veg}) = 90 \times 60 / 180 = 30$. (c) All expected frequencies are 30, 60, 30, 60; $\chi^2 = (25 - 30)^2 / 30 + (65 - 60)^2 / 60 + (35 - 30)^2 / 30 + (55 - 60)^2 / 60 = 0.833 + 0.417 + 0.833 + 0.417 = 2.5$. (d) Since $\chi^2 = 2.5 < 3.841$, do not reject H_0 : insufficient evidence of an association at the 5% level.

Quick check answer

C. $df = (\text{rows} - 1)(\text{columns} - 1) = (3 - 1)(4 - 1) = 6$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.